

Semester : 4th

Branch : Computer Engineering & IOT (TH-4(A))

Subject : IoT Architecture and Protocols (CEIPE 202 (B))

4th Semester, Computer Engineering & IOT
IoT Architecture and Protocols (CEIPE 202 (B))

UNIT – I
INTRODUCTION TO IOT

2 MARKS QUESTIONS WITH ANSWERS

Q.No.		Marking Scheme
1	Define IOT.	2
Ans.	Internet of Things (IoT) is a network of physical devices embedded with sensors, software, and connectivity to collect and exchange data over the internet without human intervention. Examples include smart thermostats and wearables.	
2	Mention any two advantages of IOT.	2
Ans.	Advantages of IoT – • Automation boosts efficiency and saves time. • Real-time monitoring enables predictive maintenance. • Data analytics improves decision-making. • Enhances safety in homes and industries. • Optimizes resources in agriculture/smart cities. (any 2)	
3	Define IIoT.	2
Ans.	IIoT stands for the Industrial Internet of Things . It refers to the use of interconnected smart sensors, actuators, and devices within industrial applications and infrastructure to collect, monitor, exchange, and analyze data.	
4	Provide a short overview of two IoT applications.	2
Ans.	Applications of IoT – • Smart Home:- Devices like smart thermostats, lights, and security cameras that can be controlled remotely. • Healthcare Monitoring: Wearable devices (e.g., fitness bands) that track vital signs and patient health metrics in real-time. • Industrial Automation: IoT sensors monitor machinery in factories for predictive maintenance, reducing downtime. They optimize production lines by analyzing real-time data on performance and failures. (any 2)	

Q.No.		Marking Scheme
5	Name the components of IOT.	2
Ans.	The four main components are: • Sensors/Devices • Connectivity (Network) • Data Processing (Cloud/Processor) • User Interface (UI)	
6	Define sensors. Name any two types of sensors.	1+1
Ans.	Sensors are physical devices that detect and measure changes in the physical environment (like temperature, motion, light) and convert them into electrical signals/data. Two Types: • Temperature Sensor (e.g., in ACs, refrigerators) • Motion Sensor (e.g., in security lights, automatic doors) • IR sensors • Ultrasonic sensors • Humidity sensors (any 2)	
7	Define UI In context of IOT.	2
Ans.	User Interface (UI) is the point of human-IoT system interaction where users can view the status of their IoT devices, access collected data (via dashboards, mobile apps), and send commands to control them.	
8.	State any two disadvantages of IOT.	2
Ans.	Two disadvantages of IoT: • Security vulnerabilities: IoT devices often have weak protections, making them easy targets for hackers to access networks and steal data. • Privacy concerns: Constant data collection from sensors tracks personal habits without consent, risking misuse by companies or breaches.	
9	Give examples of IoT in smart cities.	2
Ans.	1. Traffic Management: Sensors optimize traffic lights to reduce congestion based on real-time vehicle data. 2. Waste Management: Smart bins signal when full, improving collection efficiency in urban areas.	
10	State two IoT use cases in industry.	2
Ans.	1. Predictive maintenance: Sensors detect machine issues early to prevent breakdowns. 2. Supply chain tracking: IoT tags monitor goods location and condition in real-time.	

5 MARKS QUESTIONS WITH ANSWERS

Q.No.		Marking Scheme
1	What are the characteristics of IOT ?	5
Ans.	Characteristics of IoT – • Connectivity: IoT devices maintain constant internet connectivity for seamless communication and real-time data exchange across networks. • Intelligence: Embedded AI algorithms allow devices to analyze data autonomously and make smart decisions independently. • Scalability: IoT architectures effortlessly scale to support billions of devices in smart cities worldwide. • Dynamic Adaptability: Systems automatically adjust operations based on real-time environmental changes and conditions. • Unique Identity: Every device features a distinct identifier like IP for precise tracking and management. • Safety: Robust encryption and protocols safeguard IoT ecosystems against cyber threats and breaches. • Interoperability: Standardized protocols enable devices from different manufacturers to integrate and collaborate smoothly. • Real-time Processing: Instant data collection and analysis deliver immediate responses in critical applications. (any 5)	
2	What are the advantages of IOT ?	5
Ans.	Advantages of IoT – • Improved Efficiency: Automates routine tasks, saving time and reducing manual effort in homes and industries. • Cost Savings: Predictive maintenance prevents breakdowns, cutting repair and downtime expenses significantly. • Real-time Monitoring: Sensors provide instant data for better oversight of operations and resources. • Enhanced Safety: Detects hazards like gas leaks early, protecting people and assets proactively.	

	• Data-driven Insights: Collects vast data for analytics, enabling smarter business decisions. • Resource Optimization: Smart energy use in lighting and HVAC cuts waste and utility bills. (any 5)	
3	What are the disadvantages of IOT ?	
Ans.	Disadvantages of IoT – • Security Risks: Devices are vulnerable to hacking and unauthorized access due to weak built-in protections. • Privacy Concerns: Constant data collection invades user privacy without clear consent or protection. • High Costs: Implementation requires expensive hardware, setup, and ongoing maintenance for businesses. • Compatibility Issues: Devices from different makers often fail to integrate seamlessly. • Network Dependency: Systems fail without stable internet, causing outages in critical operations. • Data Overload: Massive data hard to manage/analyze. • Power Consumption: Battery devices drain quickly in remote use. (any 5)	5
4	Describe the IOT components.	
Ans.	IoT systems rely on five main parts working together: sensors collect data, connectivity sends it, gateways process locally, cloud analyzes deeply, and interfaces let users control everything easily. IoT Components – 1. Sensors/Devices: These parts detect things like heat, light, or motion in the environment and send simple signals to the IoT system for monitoring. 2. Connectivity: Uses Wi-Fi, Bluetooth, or mobile networks to send sensor data quickly and safely from devices to computers or clouds. 3. IoT Gateway: Connects sensors to the cloud, changes data formats, and filters info so only important details go forward for processing.	5

	4. Data Processing: Cloud or local computers analyze sensor data with smart software to find useful info and make automatic decisions. 5. User Interface: Apps and screens on phones or computers let users see data, control devices, and get alerts from anywhere easily.	
5	Write a short note on Smart Home.	
Ans.	Smart homes integrate IoT devices via Wi-Fi networks to automate daily tasks, enhance security, save energy, and provide remote control through apps and voice assistants like Alexa. Key Features • Lighting Control: Motion sensors automatically turn lights on/off when people enter/leave rooms. • Temperature Management: Smart thermostats learn habits and adjust heating/cooling remotely via apps. • Security Systems: Cameras and locks send phone alerts; unlock doors using smartphone biometrics. • Appliance Automation: Ovens, washers start/stop on schedule to optimize energy consumption. • Centralized Control: Single app manages all devices; voice commands execute multiple actions instantly.	5
6	What are the fundamental blocks of IOT?	
Ans.	Fundamental Blocks of IoT – • Sensing Layer: Sensors and actuators detect and collect real-world data such as temperature, humidity, and motion for IoT input. • Network Layer: Protocols like Wi-Fi, Bluetooth, MQTT enable secure device connectivity and reliable data transmission to process units. • Processing Layer: Edge or cloud platforms analyze data with AI/ML, generating insights and automated decisions in real-time. • Application Layer: User interfaces like mobile apps and dashboards deliver monitoring, control, and visualization of IoT outcomes.	5

10 MARKS QUESTIONS WITH ANSWERS

Q.No.		Marking Scheme
1	Explain the applications and their use cases of IOT.	10
Ans.	IoT, or the Internet of Things, refers to the network of interconnected devices that collect, share, and analyze data via the internet to enable automation, efficiency, and informed decision-making across sectors. Its applications are vast, transforming industries by integrating sensors, software, and connectivity for real-time insights and control. Applications and their use cases – Smart Homes IoT enables automated home systems for convenience and energy management. • Remote control of thermostats, lights, and appliances via smartphone apps to optimize energy use. • Security systems with motion-detecting cameras and smart locks for real-time alerts and access control. • Voice assistants like Alexa integrating devices for hands-free operation, reducing utility costs by up to 20%. Healthcare IoT supports remote monitoring and efficient care delivery. • Wearables tracking vital signs like heart rate and glucose levels for early disease detection. • Hospital asset trackers locating equipment and smart pills ensuring medication adherence. • Fall detection sensors in elderly care homes alerting caregivers instantly. Industrial Automation IoT drives manufacturing efficiency through predictive and automated processes. • Sensors on machines for vibration and temperature monitoring to enable predictive maintenance. • Real-time production line tracking to minimize downtime and defects. • Robotic arms and AGVs (automated guided vehicles) for inventory and assembly optimization. Agriculture IoT facilitates precision farming for sustainable yields. • Soil moisture sensors automate irrigation to conserve water. • Drones and cameras for crop health monitoring and pest detection. • GPS collars on livestock for location tracking and health vitals. Smart Cities	

	IoT optimizes urban infrastructure for better living. • Traffic sensors adjust signals to reduce congestion and emissions. • Smart waste bins signaling collection needs to be cut overflowed. • Streetlights dimming automatically based on motion for energy savings. Transportation and Logistics IoT enhances mobility and supply chain reliability. • Fleet telematics for real-time vehicle location and fuel efficiency. • Cargo sensors monitor temperature and humidity to prevent spoilage. • Autonomous vehicle features collision avoidance via connected data. Retail IoT streamlines shopping and operations. • RFID tags for automated inventory tracking and restocking. • Beacons sending personalized offers to shoppers' phones. • Smart shelves alerting low stock and self-checkout kiosks. Energy Sector • IoT optimizes energy production, distribution, and consumption through smart systems. • Smart grids use sensors for real-time monitoring and balancing supply-demand to prevent outages. • Predictive maintenance on assets like wind turbines and solar panels detects faults early via vibration and temperature data, reducing downtime. Environmental Monitoring • IoT enables continuous tracking of natural conditions for sustainability. • Air quality sensors in cities measure pollutants like PM2.5 and NO2, alerting authorities for interventions. • Water and soil sensors detect contamination or moisture levels, supporting conservation and disaster prevention. Education • IoT creates interactive learning environments in schools and campuses. • Smart classrooms connect boards and devices for real-time student engagement tracking and personalized lessons. • Wearables and sensors automate attendance and analyze performance data for adaptive teaching.	
2	Define IOT. What are the advantages and disadvantages of IOT?	
Ans.	Internet of Things (IoT) is a network of physical devices embedded with sensors, software, and connectivity to collect and exchange data over the internet without human intervention. Advantages of IoT – • Improved Efficiency: Automates routine tasks, saving time and reducing manual effort in homes and industries.	2+4+4

- **Cost Savings:** Predictive maintenance prevents breakdowns, cutting repair and downtime expenses significantly.
- **Real-time Monitoring:** Sensors provide instant data for better oversight of operations and resources.
- **Enhanced Safety:** Detects hazards like gas leaks early, protecting people and assets proactively.
- **Data-driven Insights:** Collects vast data for analytics, enabling smarter business decisions.
- **Resource Optimization:** Smart energy use in lighting and HVAC cuts waste and utility bills.

Disadvantages of IoT –

- **Security Risks:** Devices are vulnerable to hacking and unauthorized access due to weak built-in protections.
- **Privacy Concerns:** Constant data collection invades user privacy without clear consent or protection.
- **High Costs:** Implementation requires expensive hardware, setup, and ongoing maintenance for businesses.
- **Compatibility Issues:** Devices from different makers often fail to integrate seamlessly.
- **Network Dependency:** Systems fail without stable internet, causing outages in critical operations.
- **Data Overload:** Massive data hard to manage/analyze.
- **Power Consumption:** Battery devices drain quickly in remote use.

UNIT – II
THE IOT ARCHITECTURAL REFERENCE MODEL AS ENABLER

2 MARKS QUESTIONS WITH ANSWERS

Q.No.		Marking Scheme
1	What is the main goal of IoT ARM?	2
Ans.	The main goal is to enable interoperability, reusability, and systematic design of IoT systems by providing common standards. It helps different IoT systems communicate with each other and makes development easier.	
2	State any two benefits of IoT ARM.	2
Ans.	(1) It enables easy comparison and benchmarking of different IoT solutions and products. (2) It facilitates interoperability across different IoT systems and platforms through standardized reference models and common design principles.	
3	What is the purpose of IoT Information Model?	2
Ans.	Information Model describes how information about Physical Entities is represented digitally through Virtual Entities. It includes attributes, metadata, and value containers that store and process data about real-world objects in the IoT system.	
4	What is IoT Communication Model?	2
Ans.	Communication Model identifies communication endpoints such as Users, Resources, and Devices. It defines traffic patterns like unicast and multicast and describes the underlying technologies that enable communication in heterogeneous IoT environments.	
5	Define Virtual Entity in IoT.	2
Ans.	A Virtual Entity is a digital representation of a Physical Entity in the IoT system. It contains attributes, metadata, and services that capture and store information about real-world objects for processing and decision-making.	
6	How is IoT used in logistics for tracking?	2
Ans.	IoT sensors equipped with GPS and environmental monitors provide real-time tracking of shipment location, temperature, humidity, and shock exposure during transport. This ensures regulatory compliance, prevents spoilage, and provides complete visibility of goods in transit.	
7	Briefly explain IoT-based inventory management.	2
Ans.	RFID tags and smart shelves track stock levels in real-time, automatically detect low inventory, monitor product expiry dates, and trigger restocking alerts. This minimizes stockouts, reduces waste from expired products, and improves warehouse efficiency.	
8	List two benefits of IoT in logistics.	2
Ans.	(1) Real-time shipment tracking with continuous location and environmental condition monitoring for complete supply chain visibility.	

	(2) Reduced operational costs through route optimization, predictive maintenance, fuel consumption reduction, and automated inventory management.																						
9	List two IoT applications in healthcare.																						
Ans.	(1) Wearable health trackers and smartwatches that continuously monitor fitness metrics and vital signs like heart rate, blood oxygen, and activity levels. (2) IoT-enabled temperature-controlled containers for transporting vaccines and biological samples with real-time condition monitoring.	2																					
10	What is the relationship between Domain and Information Models?																						
Ans.	Domain Model defines the entities and their relationships in the IoT system. Information Model builds upon the Domain Model by describing how information about these entities is captured, represented, stored, and processed in digital format.	2																					
11	Differentiate between IoT and M2M communication.																						
Ans.	<table border="1"> <thead> <tr> <th>Aspect</th><th>IoT Communication</th><th>M2M Communication</th></tr> </thead> <tbody> <tr> <td>Connectivity</td><td>Internet-based (WiFi, MQTT, HTTP)</td><td>Direct point-to-point (cellular, wired)</td></tr> <tr> <td>Scalability</td><td>Highly scalable (millions of devices)</td><td>Limited (specific device pairs)</td></tr> <tr> <td>Internet Need</td><td>Requires internet/cloud</td><td>Works without internet</td></tr> <tr> <td>Data Sharing</td><td>Across apps/users/analytics</td><td>Only between connected devices</td></tr> <tr> <td>Intelligence</td><td>AI-enabled decision making</td><td>Pre-programmed instructions</td></tr> <tr> <td>Scope</td><td>Broad ecosystem</td><td>Specific task-focused</td></tr> </tbody> </table> (Any 2)	Aspect	IoT Communication	M2M Communication	Connectivity	Internet-based (WiFi, MQTT, HTTP)	Direct point-to-point (cellular, wired)	Scalability	Highly scalable (millions of devices)	Limited (specific device pairs)	Internet Need	Requires internet/cloud	Works without internet	Data Sharing	Across apps/users/analytics	Only between connected devices	Intelligence	AI-enabled decision making	Pre-programmed instructions	Scope	Broad ecosystem	Specific task-focused	1+1
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12	Name the different types of layers in IOT ARM.																						
Ans.	• Application layer • Service layer • Network layer • Device layer	2																					

5 MARKS QUESTIONS WITH ANSWERS

Q.No.		Marking Scheme
1	Explain IoT Architectural Reference Model (IoT ARM) as an enabler for IoT systems. Name the 4 models of ARM.	
Ans.	• IoT ARM is a standardized framework that provides common terminology, models, and guidelines for designing IoT systems. • It helps developers build systems that can work together easily and follow common design patterns across different domains.	5

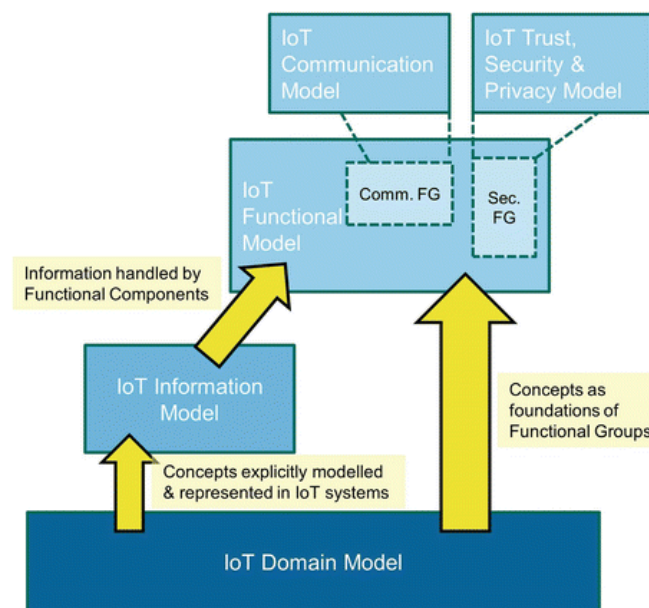
It acts as an enabler by:

- i. **Enabling interoperability** - different IoT systems can work together through common standards
- ii. **Facilitating benchmarking** - easy comparison of IoT solutions
- iii. **Supporting systematic design** - structured approach from concept to implementation
- iv. **Creating common language** - stakeholders use consistent terms
- v. **Guiding development** - helps derive concrete architectures from abstract reference models.

ARM includes four main models: Domain Model (entities and relationships), Information Model (data representation), Functional Model (component functions), and Communication Model (device interactions).

2 Describe the four main models of IoT Reference Model in brief.

IoT Reference Model consists of four interconnected models:



Ans.

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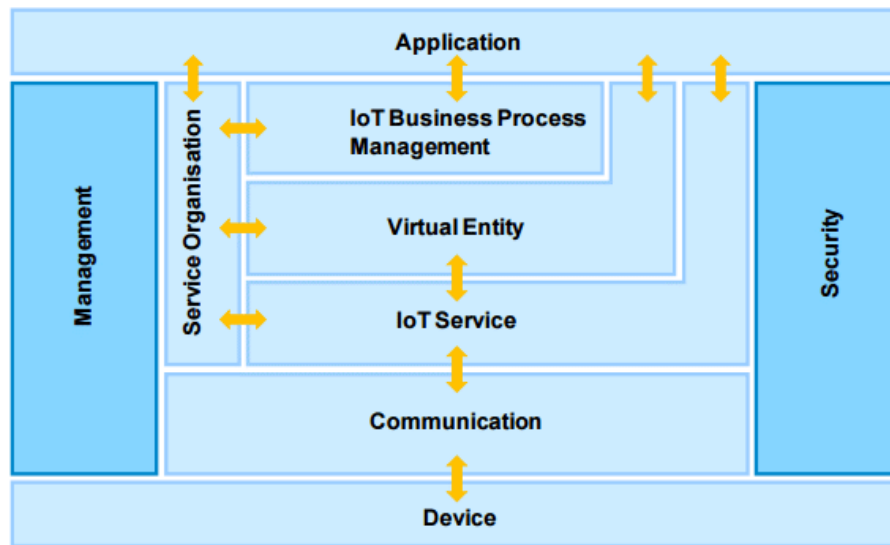
1) Domain Model

- The domain model captures the basic attributes of the main concepts and the relationship between these concepts.
- A domain model also serves as a tool for human communication between people working in the domain in question and between people who work across different domains.

2) Information Model

- The Information Model defines the structure of all IoT-related information in the Reference Model.
- This includes Virtual Entity attributes, relations, and the Semantic Grounding abstraction.

3) Functional Model



- The Functional Model describes various Functional Groups and their interactions within the IoT architecture.
- It organizes system functionality into coherent units that work together.
- The model provides a high-level view of how different functional components communicate and collaborate to enable IoT operations and services.

4) Communication Model

- The Communication Model describes communication protocols and patterns used for secure data exchange across IoT systems.
- It addresses safety, privacy protection, trust establishment, and end-to-end security mechanisms between devices, services, and users within the IoT architecture.

3 Explain IoT applications in Logistics with suitable examples.

Ans.

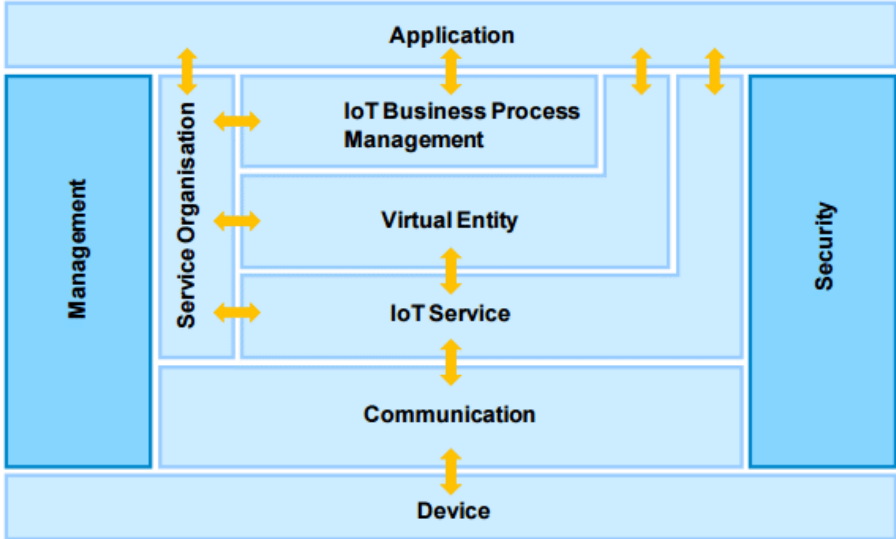
IoT transforms logistics through several applications:

1) Real-time Shipment Tracking: GPS sensors with environmental monitors track package location, temperature, humidity, and shock exposure during transport. Cloud alerts ensure regulatory compliance and prevent spoilage.

2) Cold Chain Monitoring: IoT devices in pharmaceutical containers continuously monitor temperature and humidity for vaccines/medicines, sending instant alerts when conditions deviate from safe ranges.

3) Fleet Management: Real-time vehicle tracking, predictive maintenance (detecting issues before breakdowns), driver behavior monitoring, and smart routing based on live traffic data to optimize deliveries and reduce fuel costs.

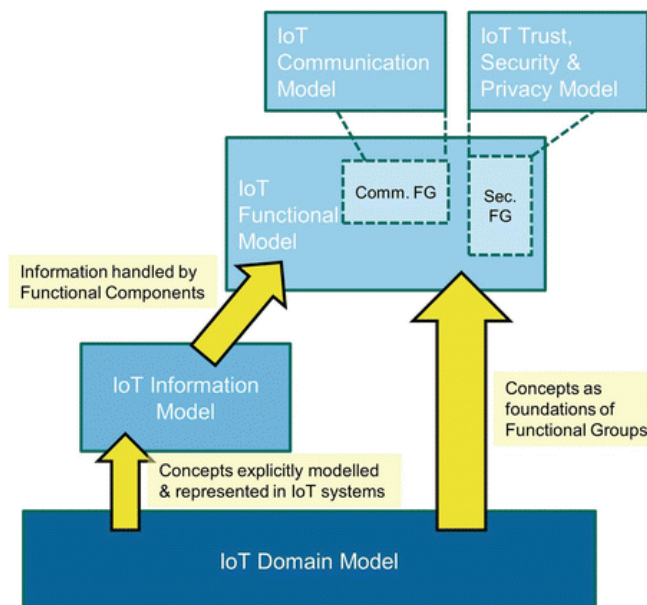
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	4) Inventory Management: RFID tags and smart shelves automatically track stock levels, monitor expiry dates, and trigger restocking alerts. Benefits include complete supply chain visibility, 30-40% cost reduction, prevented spoilage, and improved delivery efficiency. 5) Environmental monitoring - using IoT refers to continuous, real-time tracking of temperature and humidity levels throughout the supply chain, particularly for temperature-sensitive products like pharmaceuticals and perishable goods.	
4	Discuss IoT applications in the healthcare sector with practical examples.	
Ans.	IoT revolutionizes healthcare through multiple applications: 1) Remote Patient Monitoring: Wearable devices and sensors continuously monitor vital signs (blood pressure, heart rate, oxygen levels, body temperature) and automatically alert healthcare providers when abnormal conditions are detected, enabling timely medical intervention without hospital visits. 2) Smart Ambulances: IoT-enabled ambulances transmit live patient data to hospitals before arrival for better preparation, track medical equipment usage, and monitor vitals during transport. 3) Automated Medication Delivery: IoT systems deliver correct medicine doses at scheduled times with adherence tracking and error reduction. 4) Cold Chain Vaccine Transport: Temperature-controlled containers with real-time monitoring ensure biological sample safety. 5) Fitness Wearables: Smartwatches track daily health metrics. Benefits include faster diagnosis, reduced hospitalizations, improved emergency response, and enhanced patient care quality.	5
5	What is IoT Functional Model? Explain different Functional Groups in detail.	
Ans.	 The diagram illustrates the IoT Functional Model as a layered architecture. At the top is the 'Application' layer. Below it is a large container box containing several functional groups: 'IoT Business Process Management' (top), 'Virtual Entity' (middle), and 'IoT Service' (bottom). To the left of this container is the 'Management' group, and to the right is the 'Security' group. Below the container box is the 'Communication' layer, and at the very bottom is the 'Device' layer. Bidirectional yellow arrows indicate interactions between the 'Application' layer and the 'IoT Business Process Management', 'Virtual Entity', and 'IoT Service' groups. Bidirectional yellow arrows also connect the 'Management' group to the 'IoT Business Process Management' and 'Virtual Entity' groups. Bidirectional yellow arrows connect the 'Security' group to the 'IoT Business Process Management' and 'Virtual Entity' groups. Bidirectional yellow arrows connect the 'IoT Service' group to the 'Communication' layer. Bidirectional yellow arrows connect the 'Communication' layer to the 'Device' layer.	5

	The IoT Functional Model describes the Functional Groups (FGs) and their interactions in an IoT system. It defines what functions the system performs and how different components work together. 4 Main Functional Groups: 1. Device FG - Contains sensing, actuation, processing, storage, and identification components hosted on Physical Devices. 2. Communication FG - Abstracts all communication mechanisms (WiFi, cellular, mesh) used to transfer information between devices and digital systems. 3. IoT Service FG - Exposes services from Resources (devices/network) and manages service discovery and resolution. 4. Management FG - Handles system-wide configuration, fault detection, and device membership management. 5. Security FG – It manages authentication, authorization, encryption, privacy protection, and trust establishment between IoT entities. 6. Application FG - It contains custom business logic and applications specific to particular domains like healthcare, logistics, or manufacturing.	
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10 MARKS QUESTIONS WITH ANSWERS

Q.No.		Marking Scheme
1	Describe the parts of IoT Architecture reference model (ARM).	10
Ans.	The IoT Architectural Reference Model (ARM) is a comprehensive framework that establishes common language, concepts, and guidelines for designing and implementing IoT systems. It consists of two main parts: the Reference Model and the Reference Architecture, each containing specific components and sub-models. ➤ IoT Reference Model The IoT Reference Model describes the domain through four essential sub-models that provide conceptual foundation for IoT systems:	



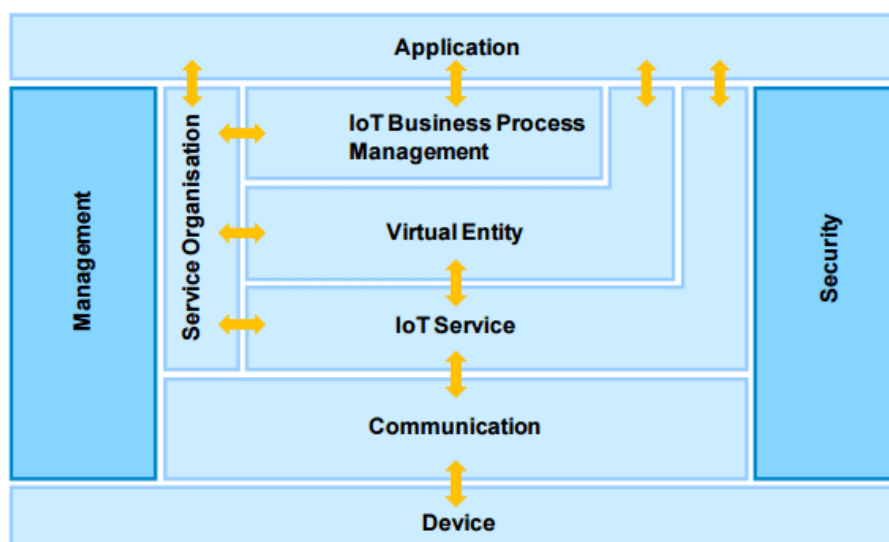
1) Domain Model

- The domain model captures the basic attributes of the main concepts and the relationship between these concepts.
- A domain model also serves as a tool for human communication between people working in the domain in question and between people who work across different domains.

2) Information Model

- The Information Model defines the structure of all IoT-related information in the Reference Model.
- This includes Virtual Entity attributes, relations, and the Semantic Grounding abstraction.

3) Functional Model



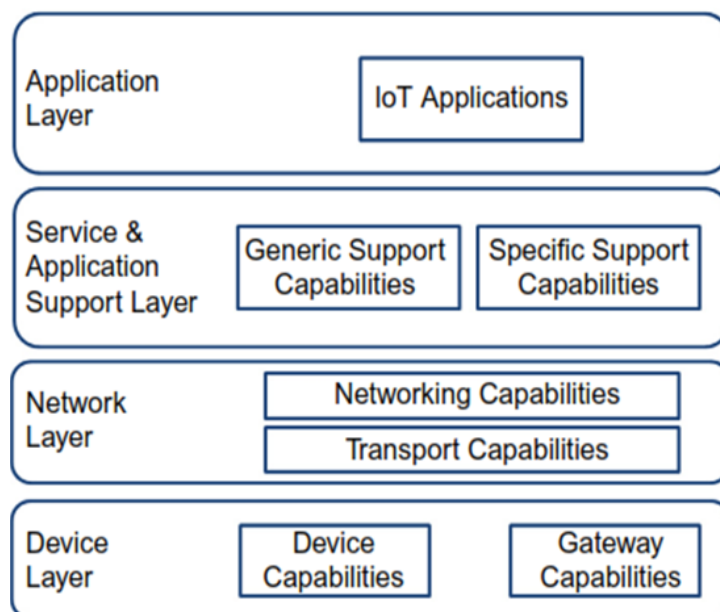
- The Functional Model describes various Functional Groups and their interactions within the IoT architecture.
- It organizes system functionality into coherent units that work together.
- The model provides a high-level view of how different functional components communicate and collaborate to enable IoT operations and services.

4) Communication Model

- The Communication Model describes communication protocols and patterns used for secure data exchange across IoT systems.
- It addresses safety, privacy protection, trust establishment, and end-to-end security mechanisms between devices, services, and users within the IoT architecture.

➤ IoT Reference Architecture

The Reference Architecture translates the conceptual Reference Model into practical, implementable system structures:



4 Main Layers –

1) Device Layer

- The Device Layer contains all physical IoT devices including sensors, actuators, and gateways that form the foundation of the IoT system.
- Device Capabilities include data sensing, data collection, processing at the edge, and device identification. Gateway Capabilities include protocol translation, network bridge functions, and connectivity between heterogeneous devices.

2) Network Layer

- The Network Layer provides communication infrastructure for devices to exchange data.

	- It contains Networking Capabilities such as connectivity protocols (WiFi, Bluetooth, Zigbee, cellular), network topology management, and data routing. 3) Service & Application Support Layer - This middle layer bridges devices and applications by providing shared functionalities. - Generic Support Capabilities include common services like data aggregation, normalization, storage, and basic analytics that all applications can use. 4) Application Layer - The Application Layer presents IoT data and services to end-users through intuitive interfaces. - It contains IoT Applications such as mobile apps, web dashboards, monitoring systems, and business intelligence tools that users interact with directly to control devices, view data, and make decisions.	
2	Explain IoT applications in Logistics and Healthcare with real examples.	
Ans.	A) IoT in Logistics Real-time Shipment Tracking - GPS and environmental sensors track location, temperature, humidity, and shock of packages. - Data is sent to a cloud platform that shows live shipment status. - Example: Amazon tracks millions of parcels globally in real time. Benefit: Accurate delivery time prediction, reduced losses and misplacement. Cold Chain Monitoring - IoT sensors placed in refrigerated trucks/containers continuously monitor temperature and humidity. - Instant alerts generated if temperature goes outside safe range. - Example: Pfizer uses IoT to maintain vaccines at around –70°C during transport. Benefit: Prevents spoilage of temperature-sensitive medicines and food, saving high costs. Fleet Management - GPS units track vehicle location; sensors monitor fuel usage and engine health. - Predictive maintenance avoids breakdowns; driver behavior (speed, harsh braking) is analyzed. - Smart routing uses traffic data for optimal paths. - Example: DHL manages 100,000+ vehicles using IoT platforms. Benefit: 20–30% fuel savings, fewer accidents, higher vehicle uptime. Inventory Management	5+5

- RFID tags and smart shelves automatically identify and count stock.
- System monitors stock levels and product expiry dates.
- Automatic purchase orders are triggered when stock falls below threshold.
- Example: Walmart uses IoT for warehouse and shelf automation.
- **Benefit:** Reduced out-of-stock cases (up to 40%), better inventory accuracy.

5. Last-mile Delivery Optimization

- Delivery routes are dynamically updated based on traffic and real-time conditions.
- Customers receive live tracking, ETA, and time window notifications.
- Supports drones and autonomous delivery vehicles.
- Example: Zomato/Swiggy show live location of delivery partners.
- **Benefit:** Faster deliveries, improved customer satisfaction, and service transparency.

B) IoT in Healthcare

1. Remote Patient Monitoring (RPM)

- Wearables (smartwatches, fitness bands) continuously measure heart rate, BP, SpO₂, temperature, etc.
- Data is sent to cloud/doctor; automatic alerts are triggered on abnormal readings.
- Example: Apple Watch detects irregular heartbeat and warns users.
- **Benefit:** Early disease detection, reduced hospital visits, continuous care at home.

2. Smart Ambulances

- Patient vitals are monitored and transmitted to hospital in real time during transit.
- Hospitals can prepare operation of theatre, ICU, and staff before patient arrival.
- GPS tracking enables optimal routing and traffic priority.
- Example: Apollo Hospitals smart ambulance systems.
- **Benefit:** Faster emergency response and improved survival rates.

3. Automated Medication Delivery / Adherence

- Smart pill dispensers remind patients of medication time and dose.
- Device ensures correct dosage and records whether medicine was taken.
- Alerts are sent to family/caregivers if a dose is missed.
- Example: Med Minder smart pill boxes.
- **Benefit:** Up to ~80% improvement in medication adherence, especially for elderly and chronic patients.

4. Hospital Asset Tracking

- RFID tags and IoT trackers are attached to equipment (wheelchairs, ventilators, infusion pumps).

	○ Real-time location and utilization data is available on a central dashboard. ○ Large hospitals can manage thousands of assets efficiently. ○ Benefit: Quick equipment location in emergencies, reduced loss/theft, better utilization. 5. Telemedicine and Remote Consultation ○ Patients consult doctors via video/audio from home. ○ Health data from home devices (BP monitor, glucometer, oximeter) is shared automatically. ○ AI can help analyze reports and flag critical cases. ○ Example: Platforms like Practo, 1mg provide online consultations. ○ Benefit: Accessible healthcare for remote/rural areas, reduced hospital crowding and travel.	
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UNIT – III
IOT REFERENCE ARCHITECTURE

2 MARKS QUESTIONS

Q.No.		Marking Scheme
1	Define IoT Reference Architecture.	2
Ans.	An IoT Reference Architecture is a standard plan or blueprint for designing and implementing IoT systems. It tells us how devices, networks, data, and applications should be arranged and how they should talk to each other so that the system is easy to build, grow, and manage.	
2	What is the functional view in IoT reference architecture?	2
Ans.	The functional view shows what the IoT system does. It lists the main groups of functions like devices, communication, services, security, and management, and explains how they work together, without worrying about where they are placed physically.	
3	What is the operation view in IoT architecture?	2
Ans.	The operation view explains how the system is run and looked after every day. It covers tasks like monitoring, logging, changing settings, fixing faults, updating software, and handling security during the whole life of the system.	
4	What do you mean by API-based architecture for IoT?	2
Ans.	In an API-based architecture, most features of the IoT system are offered through APIs, usually REST APIs over HTTP. Devices, gateways, and apps make API calls to send data, get data, or control devices. This makes integration with web and mobile apps easy.	
5	Define OPENIoT.	2
Ans.	OPENIoT is an open-source IoT platform and architecture. It helps connect sensors to the cloud, store and process sensor data, and create IoT services. It supports automatic discovery of sensors and runs many IoT tasks on cloud systems.	
6	What is meant by IoT– cloud convergence?	2
Ans.	IoT–cloud convergence means IoT devices and cloud systems work closely together. Devices collect data, and the cloud provides big storage, powerful processing, and services. Work is shared between edge devices, gateways, and cloud so the whole system is efficient.	
7	What is an API Gateway in an API-based IoT architecture?	2
Ans.	An API Gateway is a single-entry point for all API requests. It receives calls from clients, checks security, applies limits, logs requests, and forwards them to the right backend services. It hides internal details from outside users.	
8	Define service registry in SOA-based IoT architecture.	2

Ans.	A service registry is a list or directory of all available services. Each service registers its details there. Other parts of the system search this registry to find and use services at runtime.	
9	Define functional group (FG) in the IoT functional view.	2
Ans.	A functional group is a set of related functions that together give a main ability. Examples are Device FG, Communication FG, Network FG, Security FG, and Device Management FG in an IoT reference architecture.	
10	State the role of middleware in an IoT reference architecture?	2
Ans.	Middleware sits between devices and applications. It hides hardware differences, manages communication, device discovery, basic data processing, and security, and provides common services or APIs, so apps do not need to handle low-level device details.	
11	What is a microservice in an API-based IoT architecture?	2
Ans.	A microservice is a small, separate service that does one clear job, such as device registry or data ingestion. Each microservice can be deployed and scaled on its own and talks to others mainly through APIs.	
12	What is the purpose of the deployment view for an IoT designer?	2
Ans.	The deployment view helps the designer decide where to place each function (on sensor, gateway, or cloud), how devices are connected, and what delays or limits may appear. It is important for performance, cost, and reliability planning.	
13	Define QoS in an IoT reference architecture.	2
Ans.	QoS stands for Quality of Service. It means the system's promised performance, such as limits on delay, enough bandwidth, low error rate, and high availability. Different IoT applications need different QoS levels for example - real-time control needs low delay.	
14	What is the role of the scheduler in the OPENIoT architecture?	2
Ans.	In OPENIoT, the scheduler finds which sensors and data streams can satisfy a user's request. It chooses them, sets up data flows, and manages resources so that the requested IoT service runs correctly on the cloud platform.	
15	Briefly explain any two challenges of SOA in IoT.	1+1
Ans.	1. High Complexity ○ Many small services, many devices, and dynamic discovery make the system hard to design, monitor, and manage. 2. Latency (Delay) ○ Requests go through discovery, network, and middleware, so response time may become slow, which is bad for real-time IoT. 3. Security Issues ○ Many services communicating over networks increases risk of attacks, data theft, and unauthorized access. ○ Strong encryption and authentication are needed, which add overhead. (any 2)	

5 MARKS QUESTIONS

Q.No.		Marking Scheme
1	Define and explain the purpose of an IoT Reference Architecture and its key layers.	2+3
Ans.	An IoT Reference Architecture is a standard plan or blueprint for designing and implementing IoT systems. It tells us how devices, networks, data, and applications should be arranged and how they should talk to each other so that the system is easy to build, grow, and manage. It is not for one project only, but can be reused in many domains like homes, health, or industry. Data Flow 4 Stage IoT Architecture <pre> graph TD A[Application Layer Smart Applications and Management] <--> B[Data Processing Layer Processing Unit Data Analytics/ Decision Unit] B <--> C[Network Layer Internet Gateway/ Network Gateway Network Technology] C <--> D[Sensing Layer Physical Object Sensor and Actuator] </pre> Key Layers • Perception/Sensing Layer: Handles physical devices like sensors and actuators to collect environmental data and interact with the real world. • Network/Connectivity Layer: Manages data transmission via protocols (e.g., Wi-Fi, Zigbee), gateways, and networks between devices and higher layers. • Data Processing/Middleware Layer: Aggregates, stores, analyzes, and processes data using edge/cloud computing, databases, and analytics tools. • Application Layer: Delivers user-facing services like dashboards, APIs, monitoring apps, and decision-making tools based on processed insights.	

2	Describe the functional view of an IoT reference architecture with examples of functional groups.	5
Ans.	The functional view explains what parts the IoT system has and what each part does. It ignores where these parts are placed. The functional view contains information about different functional components. Related Functional Components (FCs) are grouped together in Functional Groups (FGs). Following are some of the functional groups in a functional view. • The Device FG contains sensors, actuators, and local control. • The Communication FG manages links from local networks to IP networks. • The Network FG covers routing, addressing, and firewalls. • The IoT Service FG turns device functions into services that apps can call, such as “getTemperature”. • The Security FG handles login, access control, and key management. • The Device Management FG deals with adding new devices, changing settings, and updating firmware. In a smart farm, moisture sensors (Device FG) send readings through a gateway (Communication FG) to cloud services (IoT Service FG) protected by the Security FG.	
3	What is the Information View in IoT architecture? Explain the specific parts of the data it describes.	5
Ans.	The information view is all about data. It explains what data the system creates, stores, and uses. • It covers data models, for example, how to represent a sensor reading with value, unit, and time. • It talks about semantics, that is, the meaning of data, using standard terms so different systems understand each other. • It also covers metadata like which device sent the data, at what time, and from which place. • The information view defines how data is created, cleaned, merged, averaged, stored, or deleted. • It also shows how data flows between components: who sends, who receives, and in what order.	
4	Describe the deployment view and operation view of an IoT reference architecture and explain the difference.	2+2+1
Ans.	The deployment view shows how the logical parts of the system are placed on real hardware. • It shows which functions run on sensors, gateways, edge servers, and cloud machines.	

	- It also shows network links, such as Wi-Fi, LTE, or wired links, and helps us see delays and bandwidth limits. The operation view talks about how we keep the running system healthy over time. - It includes monitoring device status, collecting logs, detecting and fixing faults, changing settings, updating software, scaling up or down, and handling security incidents. The main difference is – The deployment view answers “where and how is it installed?”, while the operation view answers “how do we run and maintain it every day?”. Both are needed for a complete design.	
5	What is SOA-based architecture for IoT? Explain its main features and advantages.	5
Ans.	IOT Service Oriented Architecture(SOA) is a design pattern that utilizes the concept of services to create a flexible and scalable IoT ecosystem. - It leverages loosely coupled services to enable the interoperability and integration of different IoT components and systems. - In IOT, SOA allows for a modular approach to connecting devices, sensors, cloud platforms, applications and user interfaces enabling, seamless communication between them. - In SOA-based IoT architecture, we think in terms of services. Each service does a clear job, like “read sensor data” or “send alert”. Main features are – Loose Coupling- Devices and applications interact via services, not direct integration Reusability- Services can be reused across multiple applications. Interoperability- Standardized service interfaces allows heterogeneous devices to work together. Scalability- New devices/services can be added without restructuring the system. Flexibility- Easy service composition and orchestration enable diverse IoT application. Advantages :- - Easier integration of different systems. - Freedom to update one service without breaking others. - Strong support for business workflows. - This is useful in large companies and projects where many back-end systems must work together.	
6	Explain API-based architecture in IoT. How is it different from traditional SOA?	2+3

Ans.	• In an API-based IoT architecture, most features are offered through simple, web-style APIs. • These are usually REST APIs over HTTP, using JSON data. • Each device, data stream, or action can have its own URL. • Clients call these URLs with GET, POST, PUT, or DELETE. <table border="1" data-bbox="264 387 1308 732"> <thead> <tr> <th>Aspect</th><th>API-based</th><th>Traditional SOA</th></tr> </thead> <tbody> <tr> <td>Protocols</td><td>REST, JSON (lightweight)</td><td>SOAP, XML (heavier)</td></tr> <tr> <td>Integration</td><td>API gateways, direct calls</td><td>Enterprise Service Bus (ESB)</td></tr> <tr> <td>Focus</td><td>Web/mobile, fine-grained reuse</td><td>Enterprise-wide services, loose coupling</td></tr> <tr> <td>Complexity</td><td>Simpler, faster</td><td>More complex, scalable for big systems</td></tr> </tbody> </table>	Aspect	API-based	Traditional SOA	Protocols	REST, JSON (lightweight)	SOAP, XML (heavier)	Integration	API gateways, direct calls	Enterprise Service Bus (ESB)	Focus	Web/mobile, fine-grained reuse	Enterprise-wide services, loose coupling	Complexity	Simpler, faster	More complex, scalable for big systems	
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7	Explain why we need different architecture views (functional, information, deployment, operation) for IoT systems.	5															
Ans.	IoT systems are complex and involve devices, networks, data, and people. One diagram cannot show everything clearly. • The functional view helps us understand what parts exist and how they work together. • The information view helps us design data formats, meanings, and flows so systems can share and analyze data correctly. • The deployment view helps us decide where to place each part in the real world and how to connect them, taking care of delays, power, and network limits. • The operation view helps us plan how to monitor, manage, and secure the running system. Different teams, like developers, data engineers, network people, and operations staff, need different views. Using all views together gives a complete and balanced design.																

Q.No.		Marking Scheme
1	Define and explain the four main architecture views of an IoT reference architecture (functional, information, deployment, and operation). How do they work together in system design?	2+5+3
Ans.	An IoT Reference Architecture is a standard plan or blueprint for designing and implementing IoT systems. It tells us how devices, networks, data, and applications should be arranged and how they should talk to each other so that the system is easy to build, grow, and manage. 1. Functional View - Shows what the system does (functions/capabilities). - Identifies main functional blocks: device layer, communication, services, data management, applications, security, management. - Describes interactions between functions (e.g., sensor → gateway → cloud service → app). - Used by designers to check that all required features are covered. 2. Information View - Focuses on data and its meaning. - Defines data models, formats, and semantics (e.g., what is “temperature reading”, units, structure). - Shows data flow between components: where data is produced, stored, processed, and consumed. - Ensures different systems can understand and use each other’s data. 3. Deployment View - Shows where components run physically. - Maps functions to nodes like sensors, gateways, edge servers, and cloud. - Considers network topology, bandwidth, latency, and hardware resources. - Helps decide placement of processing (edge vs cloud) for performance and cost. 4. Operation View - Describes how the system is operated and managed over time. - Includes monitoring, fault management, configuration, updates, and security operations.	

	- Covers device provisioning, lifecycle, and SLA/QoS management. How They Work Together Functional view defines required capabilities. Information view defines the data these functions use and exchange. Deployment view places these functions and data flows onto real hardware. Operation view ensures the deployed system can be run, monitored, and maintained reliably. Together, they give a complete blueprint for IoT system design and lifecycle.	
2	Define and explain the types and key components of SOA. What is the significance of SOA in IOT?	10
Ans.	Service Oriented IoT Architecture-Oriented IoT Architecture Service Oriented Architecture (SOA) is an architectural style where system functions are exposed as independent, reusable services with well-defined interfaces. In IoT, SOA is used to manage a large number of heterogeneous devices, sensors, and applications by abstracting their functionalities into modular services. This helps handle different protocols, platforms, and data formats in a unified way. SOA is significant in IoT because it provides: Interoperability: standard interfaces so different devices and platforms can work together. Scalability: new devices and services can be added without disturbing existing ones. Flexibility and modularity: services can be reused and recomposed for new applications. Cost efficiency: reuse of services reduces development and integration effort. Support for real-time and advanced tech: works with event-driven communication, edge computing, AI, etc. Types of SOA – Centralized SOA: a cloud hub does orchestration and analytics. Distributed SOA: edge/fog nodes handle local processing. Hybrid SOA: combines edge processing with cloud services. Event-driven SOA: services react to events like sensor triggers. Key Components of SOA in IoT 1. Service Provider	

	○ It is the party that offers a service. ○ It hosts the logic of the IoT device or application (for example: temperature reading service, location service). ○ It defines how others can call it (interface, protocol, data format). ○ 2. Service Consumer ○ It is the party that uses a service. ○ It sends a request to the provider and receives a response (for example: a mobile app that asks for sensor data). ○ It does not need to know internal details of the provider, only the interface. ○ 3. Service Registry (Directory) ○ It is like a phonebook of services. ○ Service providers register their services here with details (name, address/URL, type, protocol). ○ Service consumers search this registry to find which service to call. 4. Service Broker / Middleware ○ It works as a middle layer between providers and consumers. ○ It helps in matching a consumer request with the correct provider. ○ It can also do extra work: protocol translation, message routing, security checks, etc.	
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UNIT – IV
APPLICATION PROTOCOLS FOR IOT

2 MARKS QUESTIONS

Q. No.		Marking Scheme
1	What is UPnP in IoT?	2
Ans.	UPnP (Universal Plug and Play) is a set of networking protocols that allows devices to discover each other and communicate automatically without manual setup. It helps IoT devices like smart TVs, printers, and cameras connect easily on a home network.	
2	Define CoAP protocol.	2
Ans.	CoAP (Constrained Application Protocol) is a simple web transfer protocol designed for small IoT devices with limited power and memory. It uses UDP instead of TCP to save energy and works like HTTP but in a lighter way.	
3	What is MQTT protocol?	2
Ans.	MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol that uses a publish-subscribe model. Devices send messages to a central broker, which distributes them to interested subscribers, making it perfect for IoT communication.	
4	Give the use of XMPP in IoT.	2
Ans.	XMPP (Extensible Messaging and Presence Protocol) is used for instant messaging between IoT devices. It supports real-time communication, presence information (online/offline status), and can handle text-based data exchange between devices and servers.	
5	What is SCADA?	2
Ans.	SCADA (Supervisory Control and Data Acquisition) is a system used to monitor and control industrial equipment like power plants, water systems, and factories. It collects data from sensors and allows operators to control machines remotely.	
6	Define WebSocket protocol.	2
Ans.	WebSocket is a communication protocol that provides two-way, continuous connection between a client and server over a single TCP connection. It allows real-time data exchange without repeatedly opening new connections, ideal for live IoT dashboards.	
7	What is 6LoWPAN?	2
Ans.	6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) is a protocol that allows small IoT devices to use IPv6 addresses over low-power wireless networks. It compresses large IPv6 headers to fit in small data packets.	
8	State one feature of RPL protocol.	2
Ans.	RPL (Routing Protocol for Low-Power and Lossy Networks) creates a tree-like structure called DODAG (Destination Oriented Directed Acyclic Graph) to route data efficiently in wireless sensor networks while saving battery power.	
9	Why are authentication protocols important in IoT?	2

Ans.	Authentication protocols verify the identity of devices and users before allowing them to communicate. This prevents hackers from controlling IoT devices, stealing data, or causing damage by ensuring only trusted devices can connect.	
10	What is IEEE 802.15.4?	2
Ans.	IEEE 802.15.4 is a technical standard that defines how low-power wireless devices communicate. It specifies the physical and data link layers for short-range, low-speed wireless networks used in IoT, forming the base for ZigBee and 6LoWPAN.	
11	How is CoAP different from HTTP?	2
Ans.	CoAP uses UDP protocol (faster but less reliable) while HTTP uses TCP (slower but reliable). CoAP has smaller message sizes and lower power consumption, making it better suited for battery-powered IoT devices with limited resources.	
12	Explain publish-subscribe model in MQTT.	2
Ans.	In publish-subscribe model , devices (publishers) send messages to topics on a broker. Other devices (subscribers) receive messages from topics they are interested in. The broker handles message distribution, so devices don't communicate directly.	
13	How does WebSocket differ from HTTP?	2
Ans.	HTTP works on request-response model where client asks and server answers, then connection closes. WebSocket keeps the connection open continuously, allowing both sides to send data anytime without creating new connections repeatedly.	
14	What is the role of IEEE 802.15.4 in IoT?	2
Ans.	IEEE 802.15.4 provides the basic communication rules for wireless sensor networks. It enables low-power, short-range communication between IoT devices, allowing them to form mesh networks and communicate efficiently with minimal battery drain.	
15	What does lightweight protocol mean?	2
Ans.	A lightweight protocol is one that uses minimal data, processing power, and memory. It has simple rules and small message sizes, making it ideal for IoT devices with limited battery, storage, and computing capabilities.	

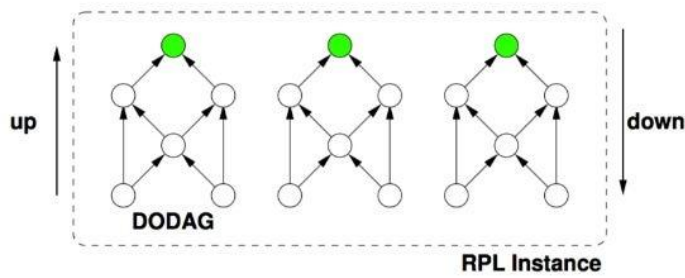
5 MARKS QUESTIONS

Q. No.		Marking Scheme
1	Explain how CoAP protocol works with an example.	5
Ans.	Constrained Application Protocol (CoAP) is an application layer protocol designed for resource-constrained devices and networks, particularly in the context of the Internet of Things (IoT). • CoAP is designed for simple devices that need to communicate using web-like methods. • It works on a client-server model where clients send requests (GET, POST, PUT, DELETE) to servers. • Since it uses UDP, messages are smaller and faster. • CoAP supports two message types: confirmable (needs acknowledgment) and non-confirmable (fire and forget). • For example, a smart temperature sensor (server) has a resource <code>/temp</code>. A mobile app (client) sends a GET request to read temperature. The sensor responds with current temperature data. • CoAP uses URI paths like <code>coap://sensor.local/temp</code> similar to HTTP URLs. Its compact 4-byte header and binary format reduce data usage, making it perfect for battery-powered sensors in smart homes or industrial monitoring.	
2	How does MQTT ensure reliable message delivery in IoT systems?	5
Ans.	MQTT (Message Queue Telemetry Transport Protocol) was designed for sending telemetry data over insecure connections. The data sending was triggered by a polling mechanism from a central server. • MQTT provides reliability through its Quality of Service (QoS) levels and broker-based architecture. • The three QoS levels are: QoS 0 (at most once - no guarantee), QoS 1 (at least once - message delivered but may duplicate), and QoS 2 (exactly once - guaranteed single delivery). • The broker stores messages and handles delivery based on QoS settings. If a subscriber is offline, the broker can retain messages for later delivery. • MQTT also uses "keep alive" mechanism where clients send periodic ping messages to maintain connection. If connection drops, the broker can deliver a "last will" message to notify other devices. • This architecture ensures important sensor data reaches its destination even on unstable networks, making MQTT reliable for critical IoT applications like healthcare monitoring or industrial control.	
3	Differentiate between UPnP and XMPP protocols in IoT.	5

Ans.	<table> <tr> <th>Point/ Feature</th><th>UPnP (Universal Plug and Play)</th><th>XMPP (Extensible Messaging and Presence Protocol)</th></tr> <tr> <td>Main purpose</td><td>Automatic device discovery and configuration in local networks</td><td>Real-time messaging and presence (online/offline) information</td></tr> <tr> <td>Typical use in IoT</td><td>Helps smart devices find and connect to each other automatically in a home/office network</td><td>Helps devices/sensors send messages, alerts, and status over the internet</td></tr> <tr> <td>Network scope</td><td>Mostly local networks (home, office LAN)</td><td>Works over the internet (wide area networks)</td></tr> <tr> <td>Protocols and data format</td><td>Uses HTTP and XML for device description and control</td><td>Uses XML streams for continuous communication</td></tr> <tr> <td>Focus</td><td>Device-centric: finding, describing, and controlling devices</td><td>Message-centric: sending and receiving messages and presence information</td></tr> <tr> <td>Extensibility</td><td>Limited extensibility (mainly fixed functions related to device discovery and control)</td><td>Highly extensible: supports many extensions (XEPs) to add new features</td></tr> <tr> <td>Example</td><td>Phone automatically finds and connects to a smart printer or smart TV using UPnP</td><td>Smart sensors send alerts to a cloud server and show online/offline status using XMPP</td></tr> </table>	Point/ Feature	UPnP (Universal Plug and Play)	XMPP (Extensible Messaging and Presence Protocol)	Main purpose	Automatic device discovery and configuration in local networks	Real-time messaging and presence (online/offline) information	Typical use in IoT	Helps smart devices find and connect to each other automatically in a home/office network	Helps devices/sensors send messages, alerts, and status over the internet	Network scope	Mostly local networks (home, office LAN)	Works over the internet (wide area networks)	Protocols and data format	Uses HTTP and XML for device description and control	Uses XML streams for continuous communication	Focus	Device-centric: finding, describing, and controlling devices	Message-centric: sending and receiving messages and presence information	Extensibility	Limited extensibility (mainly fixed functions related to device discovery and control)	Highly extensible: supports many extensions (XEPs) to add new features	Example	Phone automatically finds and connects to a smart printer or smart TV using UPnP	Smart sensors send alerts to a cloud server and show online/offline status using XMPP	
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4	How does WebSocket improve real-time communication in IoT?	5																								
Ans.	• WebSocket creates a persistent, two-way connection between client and server, solving the problem of HTTP's request-response limitation. • With HTTP, the client must repeatedly ask the server for updates (polling), wasting bandwidth and battery. • WebSocket establishes one connection that stays open, allowing both sides to send data instantly whenever needed. • The connection starts with an HTTP handshake, then upgrades to WebSocket protocol. After that, data flows freely in both directions without headers overhead. • This is perfect for IoT applications needing instant updates like live vehicle tracking (GPS data streaming continuously), smart home dashboards (showing real-time sensor readings), or remote robot control (sending commands with minimal delay). • WebSocket reduces latency from seconds to milliseconds and significantly lowers power consumption compared to constantly opening new HTTP connections.																									
5	What is 6LoWPAN and why is it important for IoT?	5																								
Ans.	6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) is a protocol that allows small IoT devices to use IPv6 addresses over low-power wireless networks. It compresses large IPv6 headers to fit in small data packets. • 6LoWPAN solves a major problem: how to use standard Internet Protocol (IPv6) on tiny wireless sensors. • IPv6 addresses are large (128 bits) and packets have big headers, but IEEE 802.15.4 networks can only send small frames (127 bytes).																									

	• 6LoWPAN uses special techniques to make this work. • Header compression reduces IPv6 headers from 40 bytes to as little as 2 bytes by removing redundant information. Fragmentation breaks large packets into smaller pieces that fit in 802.15.4 frames. • This is important because it allows every sensor to have its own IP address, making them directly accessible from the internet. • Without 6LoWPAN, sensors would need special gateways and wouldn't be part of the global internet. • Now smart city sensors, agricultural monitors, and industrial devices can all communicate using standard internet protocols.	
6	Describe RPL protocol structure and its purpose.	5
Ans.	RPL stands for Routing Protocol for Low-Power and Lossy Networks (LLN). Purpose: It was created to solve one big problem: how to route data in wireless networks that are not always reliable. In wireless IoT networks, devices send information through radio links. But those links can fail. They can be blocked, weak or noisy. Traditional routing protocols don't work well in those conditions. That's why RPL was designed specifically for IoT. RPL protocol Structure • RPL creates a routing structure specifically designed for networks with battery-powered devices and unreliable connections. • It organizes nodes into a tree-like structure called DODAG (Destination Oriented Directed Acyclic Graph). One device acts as the root (usually a gateway with power supply), and other nodes connect forming branches. • RPL supports traffic in two directions: upward (from sensors toward root, called "up" routes) and downward (from root to specific sensors, called "down" routes). • Nodes choose their parent based on an objective function that considers metrics like hop count, energy remaining, or link quality. • RPL sends control messages (DIO, DAO, DIS) to build and maintain routes. It automatically handles node failures and network changes. • For example, in a smart building, hundreds of temperature sensors form a DODAG, sending data up to a central gateway without draining batteries quickly.	

RPL topology



7

Discuss the key features of IEEE 802.15.4 standard.

5

Ans.

- IEEE 802.15.4 is specifically designed for low-power, low-data-rate wireless networks.
- **Key features include:** Low power consumption allowing devices to run on batteries for years, short-range communication (10-100 meters) suitable for sensor networks, and low data rates (250 kbps maximum) adequate for sensor readings.
- It supports two network topologies: star (all devices connect to one coordinator) and peer-to-peer (devices communicate directly forming mesh networks).
- The standard defines 16 channels in 2.4 GHz band (worldwide) and additional channels in 915 MHz and 868 MHz bands.
- It uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) to prevent data collisions.
- The frame structure includes addressing for up to 65,000 devices per network.
- IEEE 802.15.4 provides the foundation for higher-level protocols like ZigBee, Thread, and 6LoWPAN, making it essential for smart homes, industrial automation, and healthcare monitoring.

8

Discuss the comparison between MQTT, HTTP and CoAP.

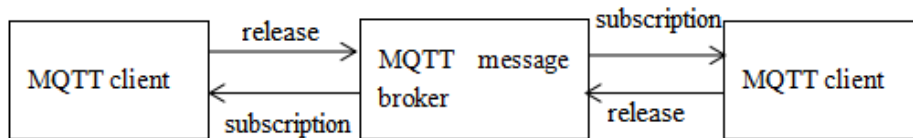
Ans.

Feature	MQTT	HTTP	CoAP
Architecture	Pub/Sub via Broker	Request/Response (Client-Server)	Request/Response (Client-Server)
Bandwidth Usage	Very low	High	Low
Protocol Stack	Runs on TCP, WebSockets	TCP/IP	UDP
Security	TLS, user auth	TLS, HTTP auth	DTLS, lightweight security

	Ideal Use Case	Real-time IoT, sensors	Web browsing, APIs	Resource-constrained IoT		
	Message Overhead	2 bytes	Large (HTTP headers)	4 bytes		

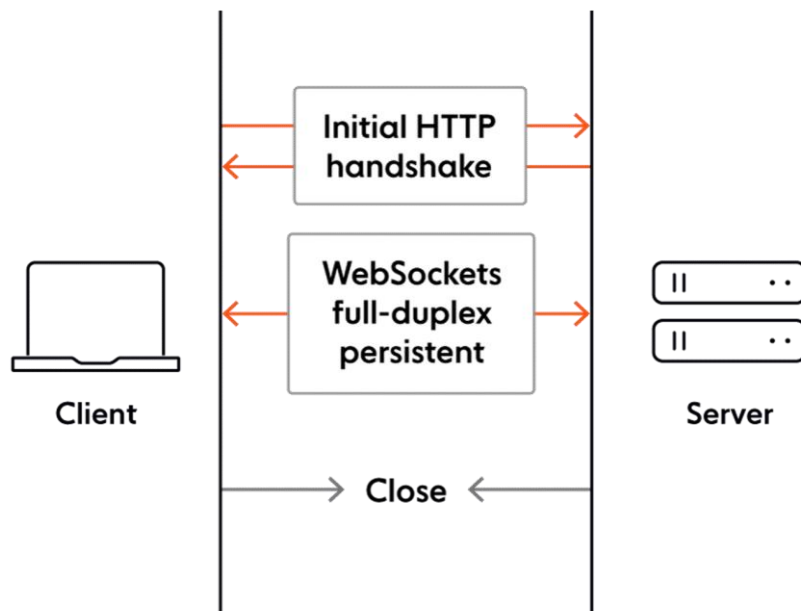
10 MARKS QUESTIONS

Q. No.		Marking Scheme
1	Compare and contrast CoAP, MQTT, and WebSocket protocols for IoT applications. Discuss their advantages and suitable use cases.	10
Ans.	CoAP, MQTT, and WebSocket are three important protocols for IoT, each with distinct characteristics. CoAP (Constrained Application Protocol): 1. Built for extremely limited devices, CoAP uses UDP making it connectionless and fast. 2. It follows REST architecture like HTTP with GET, POST, PUT, DELETE methods. 3. Messages are very small (4-byte header) and binary-encoded. 4. CoAP works well for battery-powered sensors that occasionally send readings. It supports multicasting, allowing one request to reach multiple devices. 5. However, UDP means messages might get lost requiring retransmission handling. 6. Best for: wireless sensor networks, smart agriculture, battery-operated devices. <u>CoAP</u> MQTT (Message Queuing Telemetry Transport): 1. Uses TCP for reliable delivery through a publish-subscribe model. Devices don't talk directly; they connect to a central broker. 2. MQTT header is also small (2 bytes minimum). 3. It offers three QoS levels ensuring message delivery based on importance. 4. The broker handles offline devices and message persistence. 5. MQTT is reliable but needs constant connection to broker, consuming more power than CoAP. 6. Best for: home automation, vehicle tracking, healthcare monitoring.	



WebSocket:

1. Provides full-duplex communication over TCP with persistent connection.
2. After initial HTTP handshake, the connection stays open for instant bidirectional data flow.
3. WebSocket has more overhead than CoAP or MQTT but offers real-time performance.
4. It's widely supported in web browsers making it perfect for web-based IoT dashboards.
5. Best for: real-time dashboards, live chat, remote control interfaces, web-based IoT applications.



Comparison:

- CoAP is most lightweight and power-efficient but less reliable.
- MQTT balances reliability and efficiency with broker-based architecture.
- WebSocket provides best real-time performance but higher resource usage.
- In practice, many IoT systems use multiple protocols - CoAP for sensors, MQTT for device-to-cloud communication, and WebSocket for user interfaces.

2

Describe SCADA systems in detail, including architecture, components, and how IoT integration is transforming traditional SCADA into smart industrial systems.

10

Ans.

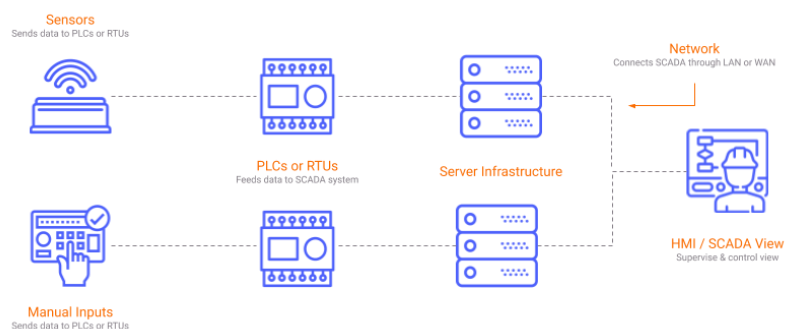
SCADA stands for "Supervisory Control and Data Acquisition". It is a control system that uses computers, networked data communications and graphical Human Machine Interfaces (HMIs) to manage and control industrial processes.

SCADA (Supervisory Control and Data Acquisition) systems are the nervous system of industrial infrastructure.

Traditional SCADA Architecture:

1. At the field level are sensors measuring parameters like temperature, pressure, flow, and level, plus actuators (valves, motors, switches) that control processes.
2. These connect to RTUs (Remote Terminal Units) - rugged devices that collect analog/digital signals and convert them to digital data, or PLCs (Programmable Logic Controllers) - small computers running control logic.
3. The communication layer traditionally used serial protocols (Modbus, DNP3) or proprietary networks to connect field devices to the control center.
4. Modern systems use Ethernet and wireless networks. At the control center, SCADA servers collect and process data, maintain databases, and run control algorithms.
5. HMI (Human Machine Interface) workstations display graphical representations of the process - like animated diagrams showing tank levels, valve positions, and alarms. Operators can view data, acknowledge alarms, and send control commands.

Simple SCADA Architecture



Components in Detail:

1. **Sensors** convert physical phenomena to electrical signals.
2. **RTUs** are intelligent devices that can perform local control even if connection to center is lost.
3. **PLCs** execute ladder logic programs for automated control sequences.
4. **Communication infrastructure** includes fiber optics, radio networks, or cellular connections.
5. **SCADA software** includes real-time database, historical trending, alarm management, and reporting tools.

IoT Transformation:

1. Modern IoT-enabled SCADA systems add revolutionary capabilities.
2. Cloud platforms store massive amounts of SCADA data and run advanced analytics.
3. Edge computing devices process data locally, reducing cloud traffic and enabling faster response.

	4. Wireless IIoT sensors are cheaper and easier to deploy than wired SCADA sensors, allowing monitoring of previously inaccessible locations. 5. Machine learning algorithms analyze historical data to predict equipment failures weeks before they happen (predictive maintenance), saving millions in downtime. 6. Mobile apps let engineers monitor and control systems from anywhere - a plant manager can check production status while traveling. 7. Digital twins create virtual replicas of physical systems for testing and optimization without risking real equipment. 8. For example, a modern smart water utility uses wireless pressure sensors throughout the distribution network (IoT sensors), PLCs control pump stations (traditional SCADA), cloud analytics detect leaks by analyzing pressure patterns (IoT analytics), and operators manage everything from a centralized dashboard accessible via mobile devices. 9. This integration transforms SCADA from a monitoring tool into an intelligent system that predicts problems, optimizes operations, and enables remote management. (Any 5 points of IOT Transformation)	
3	Discuss various authentication and security protocols used in IoT. Explain the challenges of implementing security in resource-constrained IoT devices and solutions to address them.	10
Ans.	IoT security is critical because compromised devices can cause physical harm, privacy breaches, or massive botnet attacks like the Mirai incident. Authentication Protocols: 1. Password-based authentication is simplest where devices use username/password, but IoT devices often have default passwords rarely changed by users, making them vulnerable. 2. Certificate-based authentication uses digital certificates (X.509) to verify device identity. Each device has a unique certificate issued by a trusted Certificate Authority (CA). TLS (Transport Layer Security) uses certificates to establish encrypted connections, but requires significant processing power. 3. DTLS (Datagram TLS) adapts TLS for UDP-based protocols like CoAP, providing security without TCP overhead. 4. OAuth 2.0 is token-based authentication where devices obtain temporary access tokens instead of sharing passwords repeatedly. 5. Pre-shared keys (PSK) where devices have secret keys loaded during manufacturing, used in resource-constrained devices because it's simpler than certificates. 6. Blockchain-based authentication uses distributed ledgers to verify devices without central authority, emerging in IoT applications. 7. Two-factor authentication adds extra security layer requiring multiple proofs of identity.	

Challenges in IoT Security:

1. **Limited Processing Power:** Strong encryption algorithms like RSA require significant CPU resources. A sensor with 8-bit microcontroller can't perform complex calculations quickly.
2. **Memory Constraints:** Security protocols need to store keys, certificates, and protocol state. Devices with only a few kilobytes of RAM struggle with this.
3. **Energy Limitations:** Cryptographic operations drain batteries. A sensor that should last 5 years might die in months if constantly encrypting data.
4. **Scalability:** Managing millions of device certificates and keys is administratively challenging.
5. **Lack of Updates:** Many IoT devices can't receive security patches, leaving them vulnerable forever.
6. **Physical Access:** Attackers can physically access devices installed in public places, extracting keys from memory.

Solutions and Approaches:

- **Lightweight Cryptography:** Algorithms like ECC (Elliptic Curve Cryptography) provide strong security with smaller keys (256-bit ECC equals 3072-bit RSA security). PRESENT, SPECK, and SIMON ciphers are designed specifically for constrained devices.
- **Hardware Security Modules:** Dedicated crypto chips handle encryption operations efficiently without burdening the main processor. Secure elements store keys in tamper-resistant memory.
- **Secure Boot:** Ensures only authenticated firmware runs on devices, preventing malware installation.
- **Over-the-Air Updates:** Remote update mechanisms allow patching vulnerabilities without physical access.
- **Network Segmentation:** Isolating IoT devices on separate network segments limits damage if one device is compromised.
- **Anomaly Detection:** AI systems monitor device behavior and detect unusual patterns indicating compromise.
- For example, a smart home system might use: lightweight DTLS for sensor-to-gateway communication with PSK authentication, OAuth for user smartphone access, regular automatic security updates, and network firewall isolating smart devices from computers.

UNIT – V
**CASE STUDY: CLOUD-BASED SMART-FACILITIES MANAGEMENT, HEALTHCARE,
 ENVIRONMENT MONITORING SYSTEM**

2 MARKS QUESTIONS

Q No.		Marking Scheme
1	What is Smart Facilities Management (SFM)?	2
Ans.	Smart Facilities Management (SFM) is a technology-driven system that uses IoT sensors, AI, and cloud platforms to automate and optimize building operations. It provides real-time monitoring of energy, equipment, occupancy, and environmental conditions, enabling facility managers to make data-driven decisions and improve efficiency.	
2	Define IoT-based environmental monitoring.	2
Ans.	IoT-based environmental monitoring is the systematic collection of environmental data through sensors and connected devices. • These sensors continuously gather real-time information on air quality, temperature, humidity, water quality, and other parameters. • Data is transmitted to cloud platforms for analysis, enabling immediate detection of environmental anomalies and informed decision-making.	
3	What is LoRaWAN technology in IoT applications?	2
Ans.	LoRaWAN is a low-power, wide-area networking protocol ideal for IoT applications in large facilities. It enables long-range wireless communication with minimal power consumption, allowing sensors to operate for up to 15 years on battery. LoRaWAN provides excellent indoor penetration, high reliability, and 128-bit AES encryption for secure data transmission.	
4	Give any four of the main components of Smart Facility Management systems.	2
Ans.	• IoT Sensors - Collect real-time data on temperature, humidity, occupancy, energy usage • Communication Networks - LoRaWAN/Wi-Fi transmit sensor data to cloud platforms • Cloud Platforms - Store, process, and analyze massive data volumes centrally • AI Algorithms - Provide predictive analytics and anomaly detection • Automated Controls - Adjust HVAC, lighting based on real-time conditions	

	• Dashboards - Enable remote monitoring and decision-making from anywhere (any 4)	
5	What are the benefits of cloud-based data analytics in facility management?	2
Ans.	• Centralized data from all facility sensors and systems • Real-time monitoring accessible from anywhere via internet • High scalability - Easily handles growing data volumes • Historical data preservation for trend analysis and reporting • Web/mobile dashboards for instant access and control • Rapid issue response through real-time alerts • Actionable insights for energy optimization • Predictive maintenance capabilities using AI analytics (any 4)	
6	Define predictive maintenance in smart facilities.	2
Ans.	Predictive maintenance uses IoT data and AI algorithms to forecast equipment failures before they occur. By analyzing equipment condition and performance patterns, facility managers can schedule maintenance proactively, reducing unexpected downtime, extending asset lifespan, and significantly lowering repair costs compared to reactive or scheduled maintenance approaches.	
7	Give any 2 environmental parameters which are monitored in healthcare IoT systems.	2
Ans.	• Temperature monitoring - Ensures proper vaccine/medicine storage (2-8°C) • Humidity control - Maintains optimal conditions for patient recovery • Air quality - Tracks CO₂, VOCs, particulate matter (PM2.5) levels • Air pressure - Critical for isolation rooms and clean rooms • Energy consumption - Monitors facility power usage efficiency • Oxygen levels - Ensures adequate ventilation in patient areas (any 2)	
8	How does IoT improve patient safety in hospitals?	2
Ans.	IoT sensors continuously monitor environmental conditions like temperature and humidity in critical areas (operating rooms, pharmacies, patient rooms). Real-time monitoring ensures safe vaccine storage, maintains optimal recovery conditions, supports infection control protocols, triggers alerts for abnormalities, and helps hospitals meet regulatory compliance standards, thereby protecting patients and staff.	

9	What is precision farming in agriculture using IoT?	2
Ans.	Precision farming uses IoT sensors to track soil moisture, temperature, nutrients, and weather conditions. This enables farmers to make informed decisions about irrigation, fertilization, and pest management. The method optimizes crop yields, reduces resource wastage (water, energy), and minimizes farming's environmental footprint through data-driven agriculture practices.	
10	Give the key applications of IoT in environmental monitoring.	2
Ans.	• Air Quality Monitoring Measures pollutants, CO (carbon monoxide), methane, PM2.5, NOx, and VOCs in real-time • Water Quality Monitoring Tracks pH levels, dissolved oxygen, turbidity, conductivity, temperature, and contaminants • Wastewater Management Tracks treatment efficiency, discharge quality, pH, oxygen levels in industrial and municipal systems • Deforestation Prevention Satellite and ground sensors detect illegal logging through vibration, sound, and vegetation cover changes • Wildlife Habitat Monitoring Tracks animal movement, habitat conditions, temperature, humidity for conservation planning (any 2)	
11	Briefly explain IoT sensors used in water quality monitoring?	2
Ans.	IoT sensors for water quality measure parameters like temperature, pH levels, dissolved oxygen, turbidity, conductivity, and contaminant levels. These sensors are deployed in buildings, water treatment plants, irrigation systems, and industrial processes. They enable real-time water analysis, contamination control, and conservation of this vital resource.	
12	What is occupancy monitoring in smart buildings?	2
Ans.	Occupancy monitoring uses IoT sensors to track real-time space utilization and people count. Data helps optimize HVAC settings, lighting, and workspace allocation.	

5 MARKS QUESTIONS

Q No.		Marking Scheme
1	Explain the architecture and working of a Cloud-Based Smart Facilities Management System. - A Cloud-Based Smart Facilities Management System architecture consists of four layers: Sensing Layer - IoT sensors (temperature, humidity, occupancy, energy meters) collect real-time data throughout the facility. Communication Layer - LoRaWAN, Wi-Fi, or BLE protocols transmit sensor data to gateways. Cloud Platform Layer - Central servers receive, store, and process data using AI algorithms for analytics and insights. Application Layer - Web/mobile dashboards display data, enable remote control, and send alerts. Working Process: - Sensors check building conditions like temperature and air quality. - Data goes wirelessly to gateways using Wi-Fi or LoRaWAN. - Gateways send clean data to the cloud. - AI finds patterns and problems in the data. - Managers see live updates on dashboards. - Systems auto-adjust lights or AC as needed. - Alerts warn of equipment issues early. - Past data helps improve operations over time.	5
Ans.	Example of an IoT system KADOC TECHNAMES. ALL RIGHTS RESERVED.	
2	Describe the use cases of Smart Facility Management in IoT.	5

Ans.	1. Energy Management • Automatic control of lights, fans, and HVAC based on occupancy and time. • Smart meters track power usage by floor/room to reduce electricity bills. 2. Space and Room Management • Sensors detect which rooms are used or empty. • Data helps in timetable planning, room booking, and avoiding overcrowding. 3. Security and Access Control • Smart locks, door sensors, and CCTV integration for real-time monitoring. • Alerts for unauthorized access, doors left open, or movement at night. 4. Maintenance and Asset Management • IoT sensors monitor equipment health (AC, lifts, pumps). • System creates tickets for faults and supports preventive/predictive maintenance. 5. Environment and Comfort Monitoring • Monitoring temperature, humidity, air quality, and lighting levels. • Automatic adjustments to maintain comfort and healthy indoor environment.	
4	Explain energy management and optimization in IoT-enabled smart buildings.	5
Ans.	• IoT-enabled energy management transforms buildings into intelligent, self-optimizing systems. Components: • Smart Meters - Monitor real-time electricity, gas, and water consumption at device and zone levels. • Occupancy Sensors - Detect room usage to control lighting and HVAC automatically. • Environmental Sensors - Track temperature, humidity, and natural light to optimize HVAC and lighting systems. • AI Analytics - Identify energy waste patterns and recommend optimizations. • Optimization Strategies: Automated scheduling turns off systems in unoccupied areas, saving 20-30% energy. ○ Demand response reduces consumption during peak hours when electricity is expensive. ○ Integration with solar panels and battery storage maximizes renewable energy usage. ○ Fault Detection & Diagnostics (FDD) identifies inefficient equipment early. • Results: Organizations achieve 15-40% energy cost reduction, reduce carbon footprint, and gain ESG compliance for sustainability reporting.	

5.	What are the key IoT sensors used in Smart Facility Management and their functions?	5
Ans.	Smart Facility Management relies on diverse IoT sensors for comprehensive monitoring – • Temperature & Humidity Sensors: Monitor indoor climate for optimal comfort; detect HVAC failures; essential for data centers and storage. • Occupancy Sensors: PIR/ultrasonic/camera-based detection enables automated lighting/HVAC control based on presence. • Energy Meters: Real-time tracking of electricity, gas, water usage; identifies waste and supports demand management. • Air Quality Sensors: Measure CO₂, VOCs, PM2.5, pollutants; improves health, productivity in occupied spaces. • Vibration & Acoustic Sensors: Monitor equipment health; detect abnormal patterns indicating potential failures. • Water Leak Sensors: Detect moisture/leaks in plumbing; prevents water damage and resource waste. • Light Sensors: Measure ambient light levels; optimizes artificial lighting to save energy. • Door/Window Sensors: Monitor access/security; prevents energy loss from open doors/windows. (any 5)	
6	A college has high energy bills and poor monitoring. Design a cloud based smart-facilities solution for them.	5
Ans.	Issues: • Unauthorized Access: Hackers may try to access dashboards and control building systems like doors or HVAC. • Data Theft: Energy usage, occupancy patterns, and camera data can reveal sensitive information about people and operations. • Device Attacks: Weakly protected IoT devices can be taken over and used in larger cyberattacks. • Privacy Concerns: Continuous monitoring of spaces may worry staff and students if not handled carefully. Counter measures: • Strong Authentication: Use strong passwords, multi-factor authentication, and role-based access control. • Encryption: Encrypt data in transit (HTTPS, VPN) and at rest in the cloud. • Regular Updates: Keep device firmware and cloud software updated with security patches. • Network Segmentation: Isolate IoT devices from core IT systems to limit damage if one part is hacked. • Clear Policies: Inform users what is monitored, why, and how data is protected to reduce privacy concerns.	

10 MARKS QUESTIONS

Q No.		Marking Scheme
1	Explain the working of an IoT-based environmental monitoring system using its four-stage architecture, and discuss its real-world applications, advantages, and key challenges.	10
Ans.	An IoT-based environmental monitoring system operates through a four-stage architecture that moves data from sensing to action. 1. Perception Layer (Sensing): Sensor nodes, often built around microcontrollers like ESP32, detect physical changes in the environment such as temperature, humidity, gas concentration, light, or soil moisture. These sensors continuously collect raw environmental data and send it to the next stage. 2. Network Layer (Connectivity): The collected data is transmitted using wireless protocols such as Wi-Fi, LoRaWAN, Zigbee, NB-IoT, or Bluetooth. The data may first go to a local gateway and then to the cloud, or in some cases it is sent directly to the cloud platform over cellular or Wi-Fi connections. 3. Middleware Layer (Processing): In this layer, the cloud or an edge server processes the raw data. Advanced analytics and machine-learning models are used to filter noise, remove outliers, detect anomalies, and convert sensor readings into meaningful insights such as trends, predictions, or risk levels. 4. Application Layer (Action): The insights are presented to users through web dashboards or mobile apps. Based on predefined rules or AI decisions, the system can also trigger automated actions—for example, sending alerts, switching on fans or pumps, adjusting irrigation valves, or changing HVAC settings to maintain comfort and safety. Real-World Applications • Smart Agriculture: Sensors monitor soil moisture, temperature, and micro-climate conditions to optimize irrigation, reduce water waste, and detect early signs of crop stress or disease. • Industrial Safety: Gas sensors that detect volatile organic compounds (VOCs), carbon monoxide (CO), or other hazardous gases are deployed in factories to identify leaks and prevent accidents. • Smart Cities: Air-quality sensors continuously measure pollutants such as PM2.5 and NO₂ in urban areas, helping authorities issue health advisories and manage traffic to reduce emissions. • Indoor Climate Control: Environmental sensors in offices or homes monitor temperature, humidity, and CO₂ levels, allowing HVAC systems to adjust automatically for better comfort and energy efficiency.	

	Advantages • Real-time Insights: Users get continuous, up-to-date information about environmental conditions, enabling quick decision-making. • Automation: Routine tasks such as irrigation, fan control, or alarm triggering are automated, reducing the need for manual intervention. • Cost-Effective Operation: The system prevents wastage of resources like water or energy, lowers maintenance costs, and reduces the need for frequent human monitoring. Challenges • Power Consumption: Many remote sensors must run for months or years on batteries, so low-power design and protocols like LoRaWAN are essential to extend battery life. • Data Accuracy: Sensors can drift, become dirty, or suffer environmental interference, so periodic calibration and maintenance are required to ensure reliable readings. • Scalability: Managing thousands of sensor nodes across large areas demands robust network infrastructure, efficient data management, and strong security measures to keep the system stable and secure.	
2	Analyze the role of IoT in Healthcare Environment Monitoring, covering patient monitoring, facility management, and the integration of various IoT devices in hospital ecosystems.	10
Ans.	IoT transforms healthcare by creating intelligent hospital ecosystems that improve patient care, optimize resources, and ensure safety and compliance. 1. Remote Vital Signs Monitoring: Wearable IoT devices monitor heart rate, BP, temperature, SpO₂, and ECG in real time; data sent via BLE/Wi-Fi enables remote physician supervision with 95% accuracy. 2. Chronic Disease Management: Home monitoring for COVID-19, diabetes, COPD, and cardiac patients detects health issues early, reducing hospital readmissions by 25–35%. 3. Emergency Alert Systems: IoT sensors send automatic alerts and location data to doctors when vital signs cross safe limits, supporting rapid response in critical cases. 4. Telemedicine Integration: Real-time IoT health data enhances virtual consultations, improving access for rural and underserved populations. 5. Facility Environment Monitoring: IoT sensors regulate temperature (2–8°C for vaccines, 20–24°C for operating rooms) and humidity (30–60%), ensuring safety and infection control. 6. Air and Pressure Management: Sensors maintain air quality and pressure differentials in ICUs and clean rooms, reducing hospital-acquired infections by 15–20%.	

7. Energy and Asset Management:

Smart energy systems cut consumption by 20–30%; RFID-based asset tracking reduces equipment search time by 60% and prevents losses.

8. Medication and Infection Control:

Smart refrigerators track medicine storage and expiry; IoT hygiene monitoring systems ensure hand sanitization compliance.

IoT-driven smart hospitals achieve better patient outcomes, cost savings, and operational efficiency—making IoT a cornerstone of modern healthcare systems.